

# Master Data Scientist Interview Question Bank

## 1. Python Fundamentals

### Core Python

1. What are Python's main features?
2. Difference between list, tuple, set, and dictionary?
3. Mutable vs immutable objects?
4. What happens internally when you create a Python variable?
5. is vs ==?
6. Shallow copy vs deep copy?
7. What is Python's garbage collection?
8. What is reference counting?
9. What are namespaces?
10. Local vs global vs nonlocal variables?
11. What are \*args and \*\*kwargs?
12. What are lambda functions?
13. List comprehension vs generator expression?
14. What is an iterator?
15. What is a generator?
16. yield vs return?
17. What are decorators?
18. What are closures?
19. What is a context manager?
20. Explain with open(...).
21. What are dunder methods?
22. \_\_str\_\_ vs \_\_repr\_\_?
23. What is duck typing?
24. What is monkey patching?
25. What is method resolution order (MRO)?

## **Python OOP**

26. Class vs object?
  27. Instance variable vs class variable?
  28. Instance method vs class method vs static method?
  29. Explain encapsulation.
  30. Explain abstraction.
  31. Explain inheritance.
  32. Single vs multiple vs multilevel inheritance?
  33. What does super() do?
  34. What is polymorphism?
  35. Abstract class vs interface-like design in Python?
  36. How do you implement private attributes?
  37. What is property/decorator?
  38. Composition vs inheritance?
  39. When would you use composition?
  40. Explain MRO with multiple inheritance.
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## **2. Python Coding for Data Science**

41. Reverse a string without using slicing.
42. Find duplicate elements in a list.
43. Find the first non-repeating character.
44. Count frequency of elements.
45. Find the second-largest number.
46. Find missing numbers from a sequence.
47. Remove duplicates while preserving order.
48. Flatten a nested list.
49. Merge two dictionaries.
50. Sort a dictionary by values.

51. Find common elements between two lists.
  52. Implement your own map().
  53. Implement your own filter().
  54. Implement your own reduce().
  55. Write Fibonacci using recursion.
  56. Write Fibonacci iteratively.
  57. Check whether a number is prime.
  58. Find factorial.
  59. Find palindrome.
  60. Find anagrams.
  61. Find top-K frequent elements.
  62. Implement binary search.
  63. Implement stack using Python.
  64. Implement queue using Python.
  65. Find time complexity of common Python operations.
  66. Optimize nested loops.
  67. How would you process a 20 GB CSV?
  68. How would you reduce Python memory usage?
  69. Why can vectorized NumPy code be faster than Python loops?
  70. Multiprocessing vs multithreading?
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### **3. NumPy**

71. What is NumPy?
72. NumPy array vs Python list?
73. What is vectorization?
74. What is broadcasting?
75. Explain broadcasting with an example.
76. reshape() vs resize()?

- 77. `flatten()` vs `ravel()`?
  - 78. `view()` vs `copy()`?
  - 79. What is axis 0?
  - 80. What is axis 1?
  - 81. Explain matrix multiplication.
  - 82. `*` vs `@`?
  - 83. How does NumPy handle missing values?
  - 84. What is NumPy memory layout?
  - 85. Why are NumPy operations faster?
  - 86. How would you calculate covariance using NumPy?
  - 87. How would you normalize an array?
  - 88. How would you calculate pairwise distances?
  - 89. How would you detect NaN values?
  - 90. How would you optimize a large NumPy computation?
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#### **4. Pandas**

- 91. Series vs DataFrame?
- 92. `loc` vs `iloc`?
- 93. `at` vs `iat`?
- 94. `merge()` vs `join()` vs `concat()`?
- 95. Inner vs left vs right vs outer join?
- 96. `groupby()` internally does what?
- 97. `agg()` vs `apply()` vs `transform()`?
- 98. When would you use `transform()`?
- 99. How do you handle missing values?
- 100. `dropna()` vs `fillna()`?
- 101. Mean vs median imputation?
- 102. How do you detect duplicates?

103. How do you rename columns?
  104. How do you change data types?
  105. How do you handle inconsistent categorical values?
  106. How do you detect outliers using Pandas?
  107. How do you handle dates?
  108. How do you extract year/month/day?
  109. How do you calculate rolling averages?
  110. How do you calculate cumulative values?
  111. How do you find top-N values per group?
  112. How do you identify duplicate records?
  113. How do you handle a 10 GB dataset?
  114. How do you optimize DataFrame memory?
  115. Why is apply() sometimes slow?
  116. How would you replace apply() with vectorization?
  117. How do you perform feature engineering using Pandas?
  118. How do you avoid chained assignment?
  119. What is SettingWithCopyWarning?
  120. How do you process data in chunks?
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## 5. SQL

SQL remains a major part of Data Scientist interviews, particularly joins, aggregations, window functions, and practical data manipulation.

### SQL Fundamentals

121. WHERE vs HAVING?
122. GROUP BY vs ORDER BY?
123. INNER JOIN vs LEFT JOIN?
124. UNION vs UNION ALL?
125. DELETE vs DROP vs TRUNCATE?

- 126. Primary key vs foreign key?
- 127. Candidate key vs composite key?
- 128. What is normalization?
- 129. What is denormalization?
- 130. What is an index?
- 131. Clustered vs non-clustered index?
- 132. What is a view?
- 133. What is a CTE?
- 134. Subquery vs CTE?
- 135. What is a correlated subquery?

### **SQL Interview Problems**

- 136. Find second-highest salary.
- 137. Find third-highest salary.
- 138. Find duplicate customers.
- 139. Find employees earning more than their manager.
- 140. Find top 3 salaries per department.
- 141. Find customers with no orders.
- 142. Find customers who ordered every month.
- 143. Find month-over-month revenue.
- 144. Calculate running revenue.
- 145. Calculate rolling 7-day average.
- 146. Find consecutive login days.
- 147. Find the longest customer streak.
- 148. Find first purchase per customer.
- 149. Find repeat customers.
- 150. Calculate customer retention.
- 151. Calculate churn rate using SQL.
- 152. Calculate DAU/WAU/MAU.

- 153. Calculate conversion rate.
  - 154. Find users who purchased product A but not B.
  - 155. Calculate cohort retention.
  - 156. Use ROW\_NUMBER().
  - 157. RANK() vs DENSE\_RANK().
  - 158. LEAD() vs LAG().
  - 159. What is a window function?
  - 160. How would you optimize a slow SQL query?
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## 6. Mathematics

- 161. What is a scalar?
- 162. Vector vs matrix?
- 163. Dot product?
- 164. Matrix multiplication?
- 165. Transpose?
- 166. Inverse matrix?
- 167. Determinant?
- 168. Rank of a matrix?
- 169. Eigenvalue?
- 170. Eigenvector?
- 171. Why are eigenvectors important in ML?
- 172. What is SVD?
- 173. Why is SVD useful?
- 174. What is a norm?
- 175. L1 vs L2 norm?
- 176. Euclidean distance?
- 177. Manhattan distance?
- 178. Cosine similarity?

- 179. Mahalanobis distance?
  - 180. Gradient?
  - 181. Partial derivative?
  - 182. Jacobian?
  - 183. Hessian?
  - 184. Convex vs non-convex function?
  - 185. Local vs global minimum?
  - 186. What is gradient descent?
  - 187. Why does gradient descent work?
  - 188. Learning rate too large vs too small?
  - 189. What is Taylor expansion?
  - 190. Why is calculus important in ML?
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## **7. Statistics**

- 191. Mean vs median?
- 192. When is median better than mean?
- 193. Variance?
- 194. Standard deviation?
- 195. Covariance?
- 196. Correlation?
- 197. Pearson vs Spearman correlation?
- 198. Correlation vs causation?
- 199. What is skewness?
- 200. What is kurtosis?
- 201. What is an outlier?
- 202. IQR?
- 203. Z-score?
- 204. Population vs sample?



- 205. Population variance vs sample variance?
  - 206. Standard error?
  - 207. Confidence interval?
  - 208. What does a 95% confidence interval mean?
  - 209. Central Limit Theorem?
  - 210. Law of Large Numbers?
  - 211. Sampling bias?
  - 212. Selection bias?
  - 213. Survivorship bias?
  - 214. Simpson's paradox?
  - 215. Gambler's fallacy?
  - 216. Base-rate fallacy?
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## **8. Probability**

- 217. Probability basics?
- 218. Conditional probability?
- 219. Independent vs dependent events?
- 220. Mutually exclusive events?
- 221. Bayes theorem?
- 222. Explain Bayes theorem with a medical example.
- 223. Prior vs likelihood vs posterior?
- 224. Expected value?
- 225. Variance of a random variable?
- 226. Conditional expectation?
- 227. Bernoulli distribution?
- 228. Binomial distribution?
- 229. Poisson distribution?
- 230. Normal distribution?

- 231. Uniform distribution?
  - 232. Exponential distribution?
  - 233. Gamma distribution?
  - 234. Beta distribution?
  - 235. Chi-square distribution?
  - 236. t-distribution?
  - 237. F-distribution?
  - 238. When would you use each distribution?
  - 239. PMF vs PDF vs CDF?
  - 240. Why does the normal distribution appear so often?
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## **9. Hypothesis Testing**

- 241. Null hypothesis?
- 242. Alternative hypothesis?
- 243. Type I error?
- 244. Type II error?
- 245. Statistical power?
- 246. Significance level?
- 247. p-value?
- 248. Why does  $p < 0.05$  not mean the hypothesis has a 95% probability of being true?
- 249. One-tailed vs two-tailed test?
- 250. z-test?
- 251. t-test?
- 252. paired vs independent t-test?
- 253. chi-square test?
- 254. ANOVA?
- 255. Mann-Whitney U test?

- 256. Wilcoxon test?
  - 257. What assumptions does a t-test make?
  - 258. What happens when assumptions are violated?
  - 259. Multiple hypothesis testing?
  - 260. Bonferroni correction?
  - 261. False discovery rate?
  - 262. Statistical significance vs practical significance?
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## **10. A/B Testing & Experimentation**

- 263. What is an A/B test?
- 264. How would you design an A/B experiment?
- 265. Control vs treatment?
- 266. How do you calculate sample size?
- 267. What is statistical power?
- 268. What is minimum detectable effect?
- 269. What is experiment duration?
- 270. What is randomization?
- 271. Why is randomization important?
- 272. What is sample-ratio mismatch?
- 273. What is novelty effect?
- 274. What is seasonality?
- 275. What is interference between users?
- 276. What is peeking?
- 277. Why is repeated significance testing dangerous?
- 278. Primary vs secondary metrics?
- 279. Guardrail metrics?
- 280. What if conversion increases but revenue decreases?
- 281. What if p-value is significant but effect size is tiny?

- 282. What if the experiment has insufficient power?
- 283. How would you handle multiple experiments?
- 284. How would you test a new recommendation algorithm?

Modern DS interview guides heavily emphasize experimentation and causal reasoning alongside traditional statistics.

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## **11. Machine Learning Fundamentals**

- 285. What is machine learning?
- 286. Supervised vs unsupervised learning?
- 287. Semi-supervised learning?
- 288. Self-supervised learning?
- 289. Online vs batch learning?
- 290. Parametric vs non-parametric models?
- 291. Generative vs discriminative models?
- 292. Bias vs variance?
- 293. Bias-variance tradeoff?
- 294. Underfitting?
- 295. Overfitting?
- 296. How do you detect overfitting?
- 297. How do you fix overfitting?
- 298. What is regularization?
- 299. L1 vs L2 regularization?
- 300. What is early stopping?
- 301. What is data leakage?
- 302. What is target leakage?
- 303. What is train-test contamination?
- 304. Why should preprocessing be fitted only on training data?
- 305. Why use cross-validation?

- 306. K-fold CV?
- 307. Stratified K-fold?
- 308. Group K-fold?
- 309. Time-series cross-validation?
- 310. What is a lucky split?
- 311. What is an unlucky split?
- 312. Why can repeatedly tuning against the test set be problematic?

Scikit-learn explicitly describes evaluating on unseen data and cross-validation as essential because fitting and testing on the same data can produce misleading performance.

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## **12. Regression**

- 313. Linear regression?
- 314. Derive the linear regression equation.
- 315. Ordinary Least Squares?
- 316. Cost function for linear regression?
- 317. Gradient descent vs closed-form solution?
- 318. Assumptions of linear regression?
- 319. Multicollinearity?
- 320. How do you detect multicollinearity?
- 321. VIF?
- 322. Heteroscedasticity?
- 323. Autocorrelation?
- 324. Non-normal residuals?
- 325. What happens if residuals aren't normal?
- 326. Ridge regression?
- 327. Lasso regression?
- 328. Elastic Net?
- 329. Why does L1 create sparsity?

- 330. Why doesn't ordinary L2 generally produce exact zeros?
  - 331.  $R^2$ ?
  - 332. Adjusted  $R^2$ ?
  - 333. MAE?
  - 334. MSE?
  - 335. RMSE?
  - 336. MAPE?
  - 337. When is MAPE problematic?
  - 338. How would you handle zero target values?
  - 339. How do you handle outliers in regression?
  - 340. How would you predict house prices?
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### **13. Logistic Regression**

- 341. Why is logistic regression used for classification?
- 342. Explain sigmoid.
- 343. Why not use linear regression for classification?
- 344. Log-odds?
- 345. Odds vs probability?
- 346. Logistic regression cost function?
- 347. Maximum likelihood?
- 348. Decision boundary?
- 349. What does a logistic coefficient mean?
- 350. How does regularization affect logistic regression?
- 351. Multiclass logistic regression?
- 352. One-vs-rest?
- 353. Softmax?
- 354. Logistic regression vs Naive Bayes?
- 355. Logistic regression vs decision tree?

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## 14. Decision Trees

- 356. How does a decision tree split data?
- 357. Entropy?
- 358. Information gain?
- 359. Gini impurity?
- 360. Gini vs entropy?
- 361. Why do trees overfit?
- 362. How do you control tree overfitting?
- 363. Max depth?
- 364. Min samples split?
- 365. Min samples leaf?
- 366. Pruning?
- 367. What is a greedy split?
- 368. Can decision trees handle nonlinear relationships?
- 369. Can trees handle missing values?
- 370. Are trees sensitive to feature scaling?
- 371. Why are trees interpretable?
- 372. What are limitations of decision trees?

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## 15. Ensemble Learning

- 373. What is ensemble learning?
- 374. Bagging?
- 375. Boosting?
- 376. Random Forest?
- 377. Extra Trees?
- 378. AdaBoost?
- 379. Gradient Boosting?

- 380. XGBoost?
  - 381. LightGBM?
  - 382. CatBoost?
  - 383. Random Forest vs XGBoost?
  - 384. Bagging vs boosting?
  - 385. Why does boosting work?
  - 386. Learning rate in boosting?
  - 387. Number of estimators?
  - 388. Feature importance in tree ensembles?
  - 389. Permutation importance?
  - 390. SHAP?
  - 391. Why can feature importance be misleading?
  - 392. How do you tune XGBoost?
  - 393. Why can boosting overfit?
  - 394. How do you handle class imbalance in boosting?
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## **16. SVM**

- 395. What is SVM?
- 396. What is a hyperplane?
- 397. What are support vectors?
- 398. What is margin?
- 399. Hard margin vs soft margin?
- 400. What is C?
- 401. What is the kernel trick?
- 402. Linear vs RBF kernel?
- 403. What is gamma?
- 404. C vs gamma?
- 405. Why does SVM require feature scaling?



406. SVM for nonlinear classification?

407. SVM vs logistic regression?

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## **17. KNN**

408. Why is KNN called a lazy learner?

409. How does KNN classify?

410. How do you select K?

411. What happens when K=1?

412. What happens when K is very large?

413. Why does KNN need scaling?

414. Euclidean vs Manhattan distance?

415. Curse of dimensionality?

416. KNN for regression?

417. Advantages/disadvantages of KNN?

418. How would you optimize KNN for millions of records?

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## **18. Naive Bayes**

419. What is Naive Bayes?

420. Why is it called "naive"?

421. Gaussian Naive Bayes?

422. Multinomial Naive Bayes?

423. Bernoulli Naive Bayes?

424. Why is Naive Bayes popular for NLP?

425. What is conditional independence?

426. What is the zero-frequency problem?

427. Laplace smoothing?

428. Naive Bayes vs logistic regression?

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## **19. Unsupervised Learning**

- 429. What is clustering?
- 430. K-Means?
- 431. How does K-Means work?
- 432. Why is K-Means sensitive to initialization?
- 433. K-Means++?
- 434. How do you choose K?
- 435. Elbow method?
- 436. Silhouette score?
- 437. Limitations of K-Means?
- 438. K-Medoids?
- 439. Hierarchical clustering?
- 440. Agglomerative vs divisive clustering?
- 441. Dendrogram?
- 442. DBSCAN?
- 443. What are eps and min\_samples?
- 444. Why can DBSCAN identify noise?
- 445. DBSCAN vs K-Means?
- 446. Gaussian Mixture Model?
- 447. What is EM algorithm?
- 448. Hard vs soft clustering?

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## **20. Dimensionality Reduction**

- 449. Why reduce dimensions?
- 450. Curse of dimensionality?
- 451. PCA?
- 452. Derive PCA conceptually.
- 453. Covariance matrix in PCA?

- 454. Eigenvectors in PCA?
  - 455. Principal components?
  - 456. Explained variance?
  - 457. How do you choose number of components?
  - 458. PCA vs feature selection?
  - 459. PCA before or after scaling?
  - 460. Why can PCA hurt model interpretability?
  - 461. t-SNE?
  - 462. UMAP?
  - 463. PCA vs t-SNE?
  - 464. Can t-SNE be used as a production feature transformation?
- 

## **21. Feature Engineering**

- 465. What is feature engineering?
- 466. Numerical feature transformations?
- 467. Log transformation?
- 468. Power transformation?
- 469. Box-Cox?
- 470. Yeo-Johnson?
- 471. Binning?
- 472. Interaction features?
- 473. Polynomial features?
- 474. Target encoding?
- 475. One-hot encoding?
- 476. Label encoding?
- 477. Ordinal encoding?
- 478. Frequency encoding?
- 479. When should you avoid label encoding?

- 480. Handling high-cardinality categorical variables?
  - 481. Missing-value indicators?
  - 482. Feature selection?
  - 483. Filter methods?
  - 484. Wrapper methods?
  - 485. Embedded methods?
  - 486. Recursive Feature Elimination?
  - 487. Feature leakage?
  - 488. Feature drift?
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## **22. Imbalanced Classification**

- 489. What is class imbalance?
- 490. Why can accuracy be misleading?
- 491. Precision?
- 492. Recall?
- 493. F1?
- 494. Specificity?
- 495. Sensitivity?
- 496. ROC-AUC?
- 497. PR-AUC?
- 498. Confusion matrix?
- 499. Threshold tuning?
- 500. Oversampling?
- 501. Undersampling?
- 502. SMOTE?
- 503. ADASYN?
- 504. Class weights?
- 505. Focal loss?

- 506. When should you optimize recall?
  - 507. When should you optimize precision?
  - 508. Fraud detection: which metric would you choose?
  - 509. Medical diagnosis: which metric matters?
  - 510. How do you evaluate a model with 0.1% positives?
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### **23. Probability Calibration**

- 511. What is probability calibration?
  - 512. Difference between discrimination and calibration?
  - 513. Why can a model have high AUC but poor calibration?
  - 514. Reliability diagram?
  - 515. Brier score?
  - 516. Platt scaling?
  - 517. Isotonic regression?
  - 518. When would calibrated probabilities matter?
  - 519. Why is calibration important in credit-risk models?
  - 520. How would you calibrate an XGBoost model?
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### **24. Model Evaluation**

- 521. Regression metrics?
- 522. Classification metrics?
- 523. Ranking metrics?
- 524. RMSE vs MAE?
- 525. Precision vs recall?
- 526. ROC-AUC?
- 527. PR-AUC?
- 528. Macro vs micro vs weighted F1?
- 529. Log loss?

- 530. Cross entropy?
  - 531. Confusion matrix?
  - 532. Calibration?
  - 533. Offline vs online evaluation?
  - 534. What is data leakage during evaluation?
  - 535. Why can accuracy be 99% but the model useless?
  - 536. How do you select an evaluation metric?
  - 537. Business metric vs ML metric?
  - 538. How do you evaluate a recommendation system?
  - 539. How do you evaluate a search engine?
  - 540. How do you evaluate an LLM?
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## **25. Deep Learning**

- 541. Why neural networks?
- 542. Perceptron?
- 543. Neuron?
- 544. Activation function?
- 545. ReLU?
- 546. Sigmoid?
- 547. Tanh?
- 548. Softmax?
- 549. Why does ReLU help?
- 550. Dying ReLU?
- 551. Forward propagation?
- 552. Backpropagation?
- 553. Chain rule?
- 554. Gradient descent?
- 555. Batch gradient descent?

- 556. SGD?
  - 557. Mini-batch gradient descent?
  - 558. Learning rate?
  - 559. Epoch vs iteration vs batch?
  - 560. Weight initialization?
  - 561. Xavier initialization?
  - 562. He initialization?
  - 563. Vanishing gradients?
  - 564. Exploding gradients?
  - 565. Gradient clipping?
  - 566. Batch normalization?
  - 567. Layer normalization?
  - 568. Dropout?
  - 569. Weight decay?
  - 570. Early stopping?
  - 571. Training vs validation loss?
  - 572. Why can training loss decrease while validation loss increases?
  - 573. What causes overfitting in deep learning?
  - 574. How do you debug a neural network that isn't learning?
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## **26. Optimizers**

- 575. SGD?
- 576. Momentum?
- 577. AdaGrad?
- 578. RMSProp?
- 579. Adam?
- 580. AdamW?
- 581. Adam vs AdamW?

- 582. What does momentum do?
  - 583. Learning-rate scheduling?
  - 584. Warmup?
  - 585. Cosine decay?
  - 586. One-cycle learning rate?
  - 587. Why can Adam converge faster?
  - 588. Why might SGD generalize differently from Adam?
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## **27. CNN / Computer Vision**

- 589. What is convolution?
- 590. Kernel/filter?
- 591. Stride?
- 592. Padding?
- 593. Receptive field?
- 594. Pooling?
- 595. Max pooling vs average pooling?
- 596. CNN vs fully connected network?
- 597. Why are CNNs translation-aware?
- 598. What is batch normalization in CNN?
- 599. ResNet?
- 600. Why does ResNet use skip connections?
- 601. Vanishing gradients in deep CNNs?
- 602. Transfer learning?
- 603. Data augmentation?
- 604. Object detection vs image classification?
- 605. YOLO?
- 606. Faster R-CNN?
- 607. IoU?



- 608. NMS?
  - 609. Semantic vs instance segmentation?
  - 610. U-Net?
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## **28. RNN / LSTM / GRU**

- 611. Why RNN?
  - 612. Problem with vanilla RNN?
  - 613. Vanishing gradient in RNN?
  - 614. Exploding gradient in RNN?
  - 615. LSTM?
  - 616. Forget gate?
  - 617. Input gate?
  - 618. Output gate?
  - 619. Cell state?
  - 620. GRU?
  - 621. LSTM vs GRU?
  - 622. Bidirectional RNN?
  - 623. Sequence-to-sequence architecture?
  - 624. Teacher forcing?
  - 625. Why did Transformers largely replace RNNs for NLP?
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## **29. NLP Fundamentals**

- 626. What is NLP?
- 627. Tokenization?
- 628. Sentence tokenization vs word tokenization?
- 629. Character tokenization?
- 630. Subword tokenization?
- 631. BPE?

- 632. WordPiece?
- 633. SentencePiece?
- 634. Vocabulary?
- 635. OOV problem?
- 636. Stopwords?
- 637. Stemming?
- 638. Lemmatization?
- 639. POS tagging?
- 640. Named Entity Recognition?
- 641. N-grams?
- 642. TF-IDF?
- 643. Bag of Words?
- 644. Word2Vec?
- 645. CBOW vs Skip-gram?
- 646. GloVe?
- 647. Embeddings?
- 648. Cosine similarity?
- 649. Semantic similarity?
- 650. Text classification pipeline?

Hugging Face's current documentation describes tokenizers as converting text into token IDs and producing model inputs such as attention masks; modern interviews should therefore cover tokenization and attention-mask mechanics rather than stopping at traditional NLP preprocessing.

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### **30. Transformers**

- 651. Why were Transformers introduced?
- 652. What problem does self-attention solve?
- 653. What is attention?
- 654. Query, Key, Value?

- 655. Why three projections?
- 656. Scaled dot-product attention?
- 657. Why divide by  $\sqrt{d_k}$ ?
- 658. Self-attention vs cross-attention?
- 659. Multi-head attention?
- 660. Why multiple heads?
- 661. Positional encoding?
- 662. Sinusoidal positional encoding?
- 663. Learned positional embeddings?
- 664. RoPE?
- 665. What problem does RoPE solve?
- 666. Encoder vs decoder?
- 667. Encoder-only models?
- 668. Decoder-only models?
- 669. Encoder-decoder models?
- 670. BERT?
- 671. GPT?
- 672. T5?
- 673. Causal attention?
- 674. Masked attention?
- 675. Attention mask?
- 676. Padding mask?
- 677. Causal mask?
- 678. Feed-forward network inside Transformer?
- 679. LayerNorm?
- 680. Residual connection?
- 681. Pre-LN vs Post-LN?
- 682. Transformer computational complexity?

- 683. Why is attention  $O(n^2)$ ?
  - 684. How can long-context models reduce attention cost?
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### **31. LLMs / Generative AI**

- 685. What is an LLM?
- 686. Pretraining?
- 687. Next-token prediction?
- 688. Causal language modeling?
- 689. Masked language modeling?
- 690. Parameters vs tokens?
- 691. Context window?
- 692. Temperature?
- 693. Top-k?
- 694. Top-p?
- 695. Repetition penalty?
- 696. Greedy decoding?
- 697. Beam search?
- 698. Sampling?
- 699. Hallucination?
- 700. Why do LLMs hallucinate?
- 701. Prompt engineering?
- 702. System vs user vs assistant messages?
- 703. Few-shot prompting?
- 704. Chain-of-thought?
- 705. Structured output?
- 706. Function calling?
- 707. Tool use?
- 708. Agent vs LLM?

- 709. RAG?
  - 710. RAG vs fine-tuning?
  - 711. When should you use RAG?
  - 712. When should you fine-tune?
  - 713. Embedding model?
  - 714. Vector database?
  - 715. Chunking?
  - 716. Chunk overlap?
  - 717. Semantic search?
  - 718. Hybrid search?
  - 719. Reranking?
  - 720. Retrieval evaluation?
  - 721. RAG evaluation?
  - 722. Groundedness?
  - 723. Faithfulness?
  - 724. Context precision?
  - 725. Context recall?
  - 726. LLM evaluation?
  - 727. Prompt injection?
  - 728. Data poisoning?
  - 729. LLM guardrails?
  - 730. Model quantization?
- 

## **32. LLM Fine-Tuning**

- 731. What is fine-tuning?
- 732. Pretraining vs fine-tuning?
- 733. Instruction tuning?
- 734. SFT?

- 735. LoRA?
- 736. QLoRA?
- 737. Why does LoRA reduce trainable parameters?
- 738. What is a low-rank decomposition?
- 739. What is PEFT?
- 740. Full fine-tuning vs LoRA?
- 741. What is catastrophic forgetting?
- 742. How much training data is needed?
- 743. How do you prepare instruction datasets?
- 744. How do you evaluate a fine-tuned LLM?
- 745. What causes overfitting during fine-tuning?
- 746. Learning-rate selection for LLM fine-tuning?
- 747. Gradient accumulation?
- 748. Gradient checkpointing?
- 749. Mixed precision?
- 750. BF16 vs FP16?
- 751. Quantization?
- 752. 4-bit vs 8-bit?
- 753. What is model merging?

Modern Hugging Face training workflows explicitly cover tokenization, pretrained checkpoints, learning rate, batch size, gradient accumulation, checkpointing, mixed precision and evaluation, making these practical interview topics rather than purely theoretical ones.

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### **33. RAG**

- 754. Explain a complete RAG architecture.
- 755. Document ingestion?
- 756. Document parsing?
- 757. Chunking strategies?
- 758. Fixed-size chunking?

- 759. Semantic chunking?
  - 760. Parent-child retrieval?
  - 761. Embedding generation?
  - 762. Vector indexing?
  - 763. ANN search?
  - 764. HNSW?
  - 765. FAISS?
  - 766. Metadata filtering?
  - 767. Hybrid search?
  - 768. BM25?
  - 769. Reranker?
  - 770. Top-K selection?
  - 771. Context window management?
  - 772. Query rewriting?
  - 773. Multi-query retrieval?
  - 774. HyDE?
  - 775. Retrieval failure vs generation failure?
  - 776. How would you debug a RAG system?
  - 777. How would you reduce hallucination?
  - 778. How would you evaluate retrieval?
  - 779. How would you evaluate final answers?
  - 780. How would you reduce RAG latency?
- 

### **34. AI Agents**

- 781. What is an AI agent?
- 782. Agent vs workflow?
- 783. ReAct?
- 784. Planning?

- 785. Tool calling?
  - 786. Function calling?
  - 787. Memory?
  - 788. Short-term vs long-term memory?
  - 789. Agent state?
  - 790. Multi-agent system?
  - 791. Agent orchestration?
  - 792. LangGraph?
  - 793. Why use state graphs?
  - 794. Agent loops?
  - 795. Human-in-the-loop?
  - 796. Agent evaluation?
  - 797. Tool failure handling?
  - 798. Agent hallucination?
  - 799. How do you prevent infinite agent loops?
  - 800. How would you design a production research agent?
- 

### **35. MLOps**

- 801. What is MLOps?
- 802. ML lifecycle?
- 803. Experiment tracking?
- 804. Model registry?
- 805. Model versioning?
- 806. Data versioning?
- 807. Feature store?
- 808. Training pipeline?
- 809. Inference pipeline?
- 810. Batch inference vs real-time inference?



- 811. Model serving?
  - 812. Docker?
  - 813. Kubernetes?
  - 814. CI/CD for ML?
  - 815. CI/CD vs CT?
  - 816. Model monitoring?
  - 817. Data drift?
  - 818. Concept drift?
  - 819. Prediction drift?
  - 820. Feature drift?
  - 821. Model degradation?
  - 822. Training-serving skew?
  - 823. Shadow deployment?
  - 824. Canary deployment?
  - 825. Blue-green deployment?
  - 826. A/B model deployment?
  - 827. Rollback strategy?
  - 828. Model latency?
  - 829. Throughput?
  - 830. Cost optimization?
- 

### **36. Production ML Scenarios**

- 831. Model accuracy dropped after deployment. Why?
- 832. Training accuracy is 99%, production accuracy is 70%. Diagnose.
- 833. Model works in one city but fails in another. Why?
- 834. Feature distribution changed. What do you do?
- 835. Model latency increased. How do you debug?
- 836. Prediction API suddenly returns null.

- 837. Data schema changed.
  - 838. One categorical value appears that wasn't present during training.
  - 839. Model performs well offline but poorly online.
  - 840. Model has excellent AUC but poor business performance.
  - 841. False positives increased after deployment.
  - 842. How would you monitor a fraud model?
  - 843. How would you monitor a recommendation model?
  - 844. How would you monitor an LLM application?
  - 845. How would you handle model rollback?
- 

### **37. Time Series**

- 846. What is time-series data?
- 847. Trend?
- 848. Seasonality?
- 849. Cyclicity?
- 850. Stationarity?
- 851. Why does stationarity matter?
- 852. ADF test?
- 853. Differencing?
- 854. Autocorrelation?
- 855. Partial autocorrelation?
- 856. ACF vs PACF?
- 857. White noise?
- 858. AR model?
- 859. MA model?
- 860. ARIMA?
- 861. SARIMA?
- 862. Exponential smoothing?

- 863. Prophet?
  - 864. Time-series cross-validation?
  - 865. Why can't you randomly split time-series data?
  - 866. Forecasting horizon?
  - 867. Leakage in time-series?
  - 868. How would you forecast sales?
  - 869. How would you handle missing timestamps?
  - 870. How would you handle sudden demand spikes?
- 

### **38. Recommendation Systems**

- 871. What is a recommendation system?
- 872. Content-based filtering?
- 873. Collaborative filtering?
- 874. User-item matrix?
- 875. Explicit vs implicit feedback?
- 876. Matrix factorization?
- 877. ALS?
- 878. Cold-start problem?
- 879. User cold start?
- 880. Item cold start?
- 881. Popularity baseline?
- 882. Candidate generation?
- 883. Ranking?
- 884. Two-tower model?
- 885. Embedding-based recommendation?
- 886. Precision@K?
- 887. Recall@K?
- 888. NDCG?

- 889. MAP?
  - 890. Diversity?
  - 891. Novelty?
  - 892. Exploration vs exploitation?
  - 893. How would you build Netflix-style recommendations?
- 

### **39. Causal Inference**

- 894. Correlation vs causation?
  - 895. Confounder?
  - 896. Treatment?
  - 897. Outcome?
  - 898. DAG?
  - 899. Backdoor criterion?
  - 900. Randomized controlled trial?
  - 901. Propensity score?
  - 902. Propensity score matching?
  - 903. Inverse propensity weighting?
  - 904. Treatment effect?
  - 905. Average treatment effect?
  - 906. Heterogeneous treatment effect?
  - 907. Difference-in-differences?
  - 908. Instrumental variables?
  - 909. Regression discontinuity?
  - 910. Why can observational data be misleading?
- 

### **40. Data Cleaning / Real-World Data**

- 911. How do you handle missing data?
- 912. MCAR vs MAR vs MNAR?

- 913. Mean/median/mode imputation?
  - 914. KNN imputation?
  - 915. Iterative imputation?
  - 916. Missing indicator?
  - 917. How do you detect outliers?
  - 918. Z-score vs IQR?
  - 919. Winsorization?
  - 920. Transformation?
  - 921. What if an outlier is actually a valid observation?
  - 922. Duplicate records?
  - 923. Inconsistent categories?
  - 924. Invalid dates?
  - 925. Mixed numerical/string values?
  - 926. Data type problems?
  - 927. Impossible values?
  - 928. Data validation?
  - 929. Schema validation?
  - 930. How do you build a reusable data-cleaning pipeline?
- 

#### **41. Data Leakage — Advanced**

- 931. What is data leakage?
- 932. Target leakage?
- 933. Train-test leakage?
- 934. Temporal leakage?
- 935. Feature engineering leakage?
- 936. Aggregation leakage?
- 937. Leakage through preprocessing?
- 938. Leakage through target encoding?

- 939. Leakage through oversampling?
  - 940. Leakage through feature selection?
  - 941. Give a real-world example of leakage.
  - 942. How would you detect leakage?
  - 943. Why can leakage produce 99% accuracy?
- 

## **42. Explainability**

- 944. Why explain ML models?
  - 945. Global vs local explanations?
  - 946. Feature importance?
  - 947. Permutation importance?
  - 948. SHAP?
  - 949. LIME?
  - 950. SHAP vs LIME?
  - 951. Partial dependence plot?
  - 952. ICE plot?
  - 953. What does SHAP value mean?
  - 954. Why can correlation affect SHAP interpretation?
  - 955. How would you explain a credit rejection model to a customer?
- 

## **43. ML System Design**

- 956. Design a fraud detection system.
- 957. Design a churn prediction system.
- 958. Design a recommendation system.
- 959. Design a search ranking system.
- 960. Design a credit-risk model.
- 961. Design an ad-click prediction system.
- 962. Design a demand forecasting system.

- 963. Design an image classification system.
  - 964. Design a document classification system.
  - 965. Design a RAG chatbot.
  - 966. Design an enterprise LLM assistant.
  - 967. Design a real-time prediction API.
  - 968. Design a batch ML pipeline.
  - 969. Design model monitoring.
  - 970. Design an ML feature store.
- 

#### **44. Business Case Studies**

- 971. Revenue dropped 20%. How do you investigate?
  - 972. Customer churn increased. What would you analyze?
  - 973. Conversion rate decreased. How would you debug it?
  - 974. Website traffic increased but revenue decreased. Why?
  - 975. DAU increased but retention decreased. Explain.
  - 976. How would you define success for a recommendation system?
  - 977. How would you measure a search engine?
  - 978. How would you measure customer satisfaction?
  - 979. How would you detect fraudulent transactions?
  - 980. How would you optimize marketing campaigns?
  - 981. How would you predict customer lifetime value?
  - 982. How would you segment customers?
  - 983. How would you determine why users uninstall an app?
  - 984. How would you evaluate a new product feature?
  - 985. How would you prioritize competing ML projects?
- 

#### **45. Project-Based Questions**

This is one of the **highest-value sections** for experienced candidates.

986. Explain your most recent ML project.
987. What was the business problem?
988. Why was ML required?
989. What was your baseline?
990. What data did you use?
991. How large was the dataset?
992. What were the features?
993. Which features were most important?
994. How did you handle missing values?
995. How did you handle outliers?
996. How did you handle categorical features?
997. What model did you initially choose?
998. Why did you choose that model?
999. Which models did you compare?
1000. What metric did you choose?
1001. Why that metric?
1002. What was your baseline score?
1003. What was your final score?
1004. How did you perform cross-validation?
1005. Did you encounter leakage?
1006. How did you tune hyperparameters?
1007. What was the biggest challenge?
1008. What failed?
1009. What would you change if you rebuilt it?
1010. How did you deploy it?
1011. How did you monitor it?
1012. What was the business impact?
1013. How much latency did the system have?



- 1014. How much did inference cost?
  - 1015. How did you handle model drift?
- 

#### **46. Behavioral / Senior-Level Questions**

- 1016. Tell me about yourself.
  - 1017. Explain your career journey.
  - 1018. Why Data Science?
  - 1019. Why this company?
  - 1020. Why should we hire you?
  - 1021. Tell me about your biggest ML failure.
  - 1022. Tell me about a disagreement with an engineer.
  - 1023. Tell me about a disagreement with a product manager.
  - 1024. Tell me about a project that failed.
  - 1025. Tell me about a difficult dataset.
  - 1026. Tell me about a model that didn't work.
  - 1027. Tell me about a time you influenced a business decision.
  - 1028. How do you communicate ML results to nontechnical stakeholders?
  - 1029. How do you prioritize projects?
  - 1030. How do you handle ambiguous requirements?
  - 1031. How do you decide whether ML is actually necessary?
  - 1032. What would you do if your manager asks for an unrealistic model accuracy?
  - 1033. How do you mentor junior data scientists?
  - 1034. What is the biggest mistake Data Scientists make?
  - 1035. What is the difference between a good model and a good ML product?
- 

#### **47. The Questions That Separate Junior from Senior**

These are the questions I would especially prepare for **your level**:

1. **Why did you choose this model instead of another model?**

2. **What is your baseline?**
3. **What would happen if the data distribution changed?**
4. **How do you know your model isn't leaking information?**
5. **Why is your metric appropriate for the business?**
6. **What happens when your model goes into production?**
7. **How do you monitor model performance when labels arrive weeks later?**
8. **What if offline AUC improves but revenue decreases?**
9. **How do you handle concept drift?**
10. **How would you reduce inference latency?**
11. **How would you reduce model cost?**
12. **How would you explain your model to a nontechnical stakeholder?**
13. **What happens when your assumptions are violated?**
14. **How would you design the experiment to prove causality?**
15. **How would you debug a model that suddenly degraded?**
16. **How would you decide between RAG and fine-tuning?**
17. **How would you evaluate an LLM application?**
18. **How would you evaluate retrieval separately from generation?**
19. **How would you design a production-grade ML system?**
20. **What trade-off did you make in your project?**